

QUESTIONS

LEVEL 3 — HARD

Q1

Numbers



Q2

Control



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Level 3 — Hard (H) Perfect Squares — Numbers

✓ Submitted

Write a Program to create a list of all numbers in a range which are perfect squares and the sum of the digits of the number is less than 10.

Sample Input & Output:

Enter lower range: 1

Enter upper range: 40

[1, 4, 9, 16, 25, 36]

Test case:

1. Enter lower range: 50

Enter upper range: 100

2. Enter lower range: 5

Enter upper range: 8

3. Enter lower range: 10

Enter upper range: 5

4. Enter lower range: 300

Enter upper range: 500

5. Enter lower range: 0

Enter upper range: -100

Testcase:

Your Code

```
1 import java.util.*;  
2 class main{  
3     static int sum(int n){  
4         int s=0;  
5         while(n>0){  
6             s+= n%10;  
7             n/=10;  
8         }  
9         return s;  
10    }
```

Output

```
[1, 4, 9, 16, 25, 36, 81, 100]
```

Enter upper range: 500
5. Enter lower range: 0
Enter upper range: -100
Testcase:

Your Code

```
1 import java.util.*;  
2 class main{  
3     static int sum(int n){  
4         int s=0;  
5         while(n>0){  
6             s+= n%10;  
7             n/=10;  
8         }  
9         return s;  
10    }  
11 public static void main(String[] args){  
12     Scanner sc=new Scanner(System.in);  
13     int l=sc.nextInt(), u=sc.nextInt();  
14     List<Integer> a= new ArrayList<>();  
15     for (int i=Math.max(0, l); i<=u; i++){  
16         int r=(int)Math.sqrt(i);  
17         if(r*r==i && sum(i)<10)  
18             a.add(i);  
19     }  
20     System.out.println(a);  
21 }  
22 }
```

Output

```
[1, 4, 9, 16, 25, 36, 81, 100]
```

Next →

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QUESTIONS

LEVEL 3 — HARD

Q1

Numbers



Q2

Control



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Level 3 — Hard (H) Steps — Control

✓ Submitted

Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.

Example 1:

Input: num = 14

Output: 6

Explanation:

Step 1) 14 is even; divide by 2 and obtain 7.

Step 2) 7 is odd; subtract 1 and obtain 6.

Step 3) 6 is even; divide by 2 and obtain 3.

Step 4) 3 is odd; subtract 1 and obtain 2.

Step 5) 2 is even; divide by 2 and obtain 1.

Step 6) 1 is odd; subtract 1 and obtain 0.

Testcase: 1. n = 5

2. n = 10

3. n = 12

4. n = 18

5. n = 20

Your Code

```
1 import java.util.Scanner;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int num=sc.nextInt(), steps=0;
6         while (num > 0){
7
8             if(num%2==0)
```

Output

5

Step 4) 3 is odd; subtract 1 and obtain 2.
Step 5) 2 is even; divide by 2 and obtain 1.
Step 6) 1 is odd; subtract 1 and obtain 0.
Testcase:1. n = 5
2. n = 10
3. n = 12
4. n = 18
5. n = 20

Your Code

```
1 import java.util.Scanner;  
2 class Main{  
3     public static void main(String[] args){  
4         Scanner sc = new Scanner(System.in);  
5         int num=sc.nextInt(), steps=0;  
6         while (num > 0){  
7  
8             if(num%2==0)  
9                 num/=2;  
10            else  
11                num--;  
12            steps++;  
13        }  
14        System.out.println(steps);  
15    }  
16 }
```

Output

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